



Studying with GenAI: Student views on the opportunities and risks of GenAI in higher education

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Abstract

The emergence of Generative Artificial Intelligence (GenAI) technology and the increasing sophistication of such tools have raised concerns about their impact on teaching, learning, and assessment in higher education. To gain an understanding of how students are using GenAI tools for studying and assessments, and to explore their perceptions of its benefits and challenges in higher education, this qualitative study examined the views of Psychology students from two UK universities. Six focus groups ($N=28$), consisting of postgraduate psychology students and a few UG students were conducted using semi-structured interview questions to explore their experiences and attitudes toward the use of GenAI during their studies. Students reported using GenAI to summarise content, clarify complex ideas, and ease the writing process. While they recognised its value, especially for accessibility and time management, they also expressed concerns about plagiarism, reduced critical thinking, and unclear boundaries of ethical use. Many felt anxious about academic integrity due to inconsistent guidance. Notably, some students viewed AI use as ethically problematic, particularly in group work. The findings underscore the urgent need for clearer institutional policies and targeted training to promote ethical, effective engagement with GenAI in academic contexts.

Keywords AI in higher education · GenAI and learning · Technology and learning

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1 Introduction

The emergence of Generative Artificial Intelligence (GenAI) tools such as ChatGPT, has raised challenging questions about the nature of teaching, learning, and assessments in Higher Education (HE). These GenAI tools have recently gained significant popularity in academic settings due to their capabilities of assisting with literature reviews, essay writing, research, and other academic projects (Bahroun et al., 2023; Stojanov, 2023; Strzelecki, 2023). One pertinent question is how GenAI can potentially transform the learning landscape as we know it in, HE and the impact this will have on student experiences. Pedagogical research to date offers a fusion of findings rooted in opportunities and challenges (Nanaware & Sharma, 2021) of GenAI in teaching, learning, and assessment. Opportunities relate to personalised learning according to the preferences of each student (Bahroun et al., 2023) and the ability to control iterations of an assessed topic to improve quality of their work (Akinwalere & Ivanov, 2022), with challenges related to concerns around accuracy of content, comprehensiveness of information, and potential replacement of the learning experience for students (Akinwalere & Ivanov, 2022). Despite these polarising findings, it is apparent GenAI tools will only embed deeper into the learning experiences of students in HE and inevitably change the way Higher Education works. The current study aims to explore the perspectives and experiences of students using GenAI for their learning whilst studying in Higher Education institutions.

1.1 Students' learning experiences in higher education

A recent monumental shift in teaching, learning, and assessment practices in Higher Education came with the worldwide pandemic of COVID-19 in 2020. Whilst there has been wide discussion concerning the impact of COVID-19 on teaching practices and the shift to online learning (Anderson, 2020; Cena et al., 2024; Watermeyer et al., 2021).

significantly less attention has been paid to the impact of the pandemic on student's learning experiences. Ultimately, this led to poor learning experiences for students as they navigated a unique set of challenges independently. On the other hand, recent research has also shown that online teaching, learning, and assessment in Higher Education promotes students' motivation, satisfaction, outcomes, attitudes, skills (Neji et al., 2023) and plays a mediating role in academic self-efficacy (Acosta-Enriquez et al., 2025).

In many ways, the rapid pivot to online learning during the COVID-19 pandemic paved the way for students and educators to embrace the current surge of Artificial Intelligence (AI) with its heightened experience of using adaptive systems to support learning processes, receiving online support, and seeking information in relevant search engines. Generative AI, however, has become much more advanced in the last few years and now focuses on mimicking human cognitive functions such as decision-making, problem-solving, and learning (Bellman, 1978) and is now believed to have the potential to transform the education arena than any technology before it (Ifenthaler et al., 2024). Currently, GenAI is used to assist with making data-driven decisions based on student engagement data, for example, through Virtual Learning

Environments. It has also been used to identify students at risk of dropping out or experiencing mental health issues using predictive analytics to track ‘at-risk’ behaviours and trigger alerts to educators/advisors (Herodotou et al., 2021). Similarly, GenAI has provided support with student experience with HE institutions, developing an automated digital assistant that supports students’ academic and emotional wellbeing. This GenAI tool allows users to navigate their timetable, contact members of staff, request important documents, engage with their students’ union (SU) and extracurricular activities, and much more (König et al., 2024).

1.2 GenAI and teaching, learning, and assessment practices

A recent scoping review by Hay et al. (2024) concluded that overall, GenAI added three opportunities to students’ learning experiences—perceived unbiased feedback, diverse sources of information, and self-assessment of their writing. The main challenge faced was in relation to student academic integrity (Xia et al., 2024). In terms of these opportunities, there is a prevailing consensus that GenAI exhibits satisfactory competence in generating coherent texts and feedback to students through ChatGPT (Dai et al., 2023). HE organisations, such as Advance HE, have provided tentative steps to help the process of learning and assessment redesign in light of GenAI by: (1) understanding how specific assessments allow for AI collusion, (2) mapping student learning against relevant parameters including context, format etc., and (3) redesigning learning by creating assessments that involve multidimensional tasks which encourage AI collaboration (Advance HE, 2023). Whilst these steps provide helpful suggestions, they also reinforce the need for teaching, learning, and assessment reform in GenAI, a timely task for any HE institution. The challenge highlights the concern that many students can simply bypass the learning process and submit AI-generated work. In the absence of any robust and reliable AI detection tools (Lodge et al., 2023), students can gain an unfair advantage in their assessments (Cotton et al., 2024). Already, this is providing many issues in Higher Education institutions, which were left with the pertinent question: is a student colluding if they use GenAI at any point throughout the learning process (research, planning, writing)? This ultimately led to the development of GenAI policies to provide a blueprint of acceptable and unacceptable use of GenAI. A relevant example is the GenAI policy guidelines co-designed by 24 Russell Group universities in the UK, which emphasises how universities should support students and staff to become AI-literate while attending to issues of privacy and plagiarism (Russell Group Committee, 2023). The ongoing debate of whether AI detection tools should be used in teaching, learning, and assessment practices of HE remains at a lively stage.

1.3 Adoption of using GenAI in higher education

Some important aspects of considering GenAI in HE relate to: (1) skills which educators have using the technology and (2) their ability to successfully integrate this into curricula.

Rather than speculating whether GenAI will replace academic and faculty staff in HE, it is now widely accepted that understanding the advantages that GenAI offers

educators is a more reasonable approach (Hrastinski et al., 2019). Prior to the recent growth, GenAI has long been used to allow educators to check the authenticity of written assignments submitted by students using AI-based systems (e.g. Turnitin). This has now progressed to educators using GenAI to provide feedback on assessments, with a consensus that it improves marking turnaround times (Mumford & Atay, 2021) and ensures consistency with feedback to students (Ellis et al., 2022). Indeed, one study has also demonstrated students' preference for assignment feedback that has been augmented by AI (Schultze-Gerlach et al., 2024). A recent qualitative study by Mutanga et al. (2024) explored lecturers' attitudes towards AI adoption into their learning activities and assessment and feedback. Researchers found largely mixed responses with some lecturers eager to embrace AI technologies while others are more resistant to their use in the absence of institutional/policy guidance and examples of best practice (Mutanga et al., 2024). This is corroborated in a study by Opesemowo et al. (2024), which investigated lecturers' attitudes and perceptions regarding the use of ChatGPT for instructional assessment. The study found that lecturers held low perceptions in terms of the viability of use and had poor perception levels when it came to exploring the potential for ChatGPT in their teaching and learning practices. Perhaps this is best conceptualised in terms of 'cautious optimism' amongst educators regarding the use of GenAI in their pedagogical methods (Mutanga et al., 2024).

1.4 Students' experiences of using GenAI

Whilst literature on students' experiences of implementing AI into their learning is small but growing, one study by Playfoot et al. (2024) examined the relationship between degree apathy and the willingness of students to use AI tools in their studies. Researchers found that students with higher levels of degree apathy were more inclined to misuse AI tools, highlighting a significant risk factor for academic misconduct. These findings further reinforce the appeal of GenAI tools such as ChatGPT to those students who find themselves disengaged with the programme and course content. However, AI tools can be incorporated into learning practices in a positive way if students educate themselves on the uses of GenAI and indeed the limitations. For example, students have reported reduced stress when interacting with GenAI chatbots whilst practicing new skills in healthcare delivery as this allows real-time user feedback and tailored information based on their input (Korngiebel & Mooney, 2021). Interestingly, it was also found that high-performing students were significantly more likely to utilise AI tools compared to low-performing students (Sweeney, 2023). The main purpose is to acquire additional support which is consistent with the notion that high-performing students utilise a wide range of educational technologies while lower-achieving students are more likely to employ contract cheating services to complete assignments for them (Sweeney, 2023). Indeed, high-achieving students are more likely to recognise the benefits of using AI tools to aid their expertise when completing assignments and attempting to better understand course content.

An important issue is whether students' use of AI actually *improves* their overall academic performance. Currently, available studies demonstrate that the overall effect of using GenAI is unclear, and there is no consensus on mediating factors that

influence this effect, for example, degree programme, assessment type, student ability etc. (Zhang et al., 2024). Some studies have found that students using ChatGPT excelled in their academic performance (Ji et al. 2023a, b) and that using ChatGPT enhanced the quality of essays and overall marks for college students (Bašić et al., 2023). However, a study by Bernabei et al. (2023) examined the use of AI in engineering education while students were required to use ChatGPT for an essay assignment. The researchers reported that the AI-generated submissions were of reasonably high quality, but the use of AI had limited effects on overall student performance. In a similar study by Smerdon (2024), researchers found no significant effect of AI use on research proposal marks among second-year undergraduate economics students at an Australian university. Thus, it is evident that the impact of GenAI on students' academic achievements when completing assessments has not been widely established and lacks convincing evidence.

1.5 Student perceptions and acceptance of GenAI

Another key consideration relevant to the current study is students' perceptions of using GenAI in their learning journey in HE institutions. Student perceptions of their learning environment play a significant role in their overall engagement with course content, and more generally, in their academic and career outcomes. This is highlighted in Biggs' 3P (Presage–Process–Product) model of teaching and learning (Biggs & Tang, 2011), which links positive perceptions of the learning environment (e.g., curriculum content, teaching methods, learning resources, assessment and feedback practices) to greater confidence and empowerment in students' abilities to adopt deep learning approaches.

When considered in the context of GenAI, students who perceive technological innovations positively are more likely to be willing to use these tools in their learning. A study by Dahmash et al. (2020) found that students with a good understanding of AI expressed lower levels of anxiety about its use. This highlights the importance of educating students on how to effectively use GenAI in their academic journey, enabling them to achieve more favourable outcomes. A more recent study by Chan and Hu (2023) in Hong Kong, found that students demonstrated positive attitudes toward using GenAI technologies in their learning, research, and future careers. Interestingly, these positive perceptions were positively correlated with the frequency of GenAI use. The study also identified ongoing concerns students have when using GenAI, such as issues around accuracy, plagiarism, and ethics (also noted in Dahmash et al., 2020), which are now common themes in GenAI literature.

In relation to perceptions, recent studies have also explored students' acceptance of GenAI tools through the lens of the Technology Acceptance Model (TAM; Davis, 1985). This inclusion is particularly relevant, as the TAM framework suggests that students' perceptions of a technology's usefulness and ease of use are key determinants of their intention to adopt it. A study by Almassaad et al. (2024) revealed that 78.7% of higher education students (from a sample of 859 in Saudi Arabia) frequently used GenAI tools, primarily for defining and clarifying concepts, translation, generating writing ideas, and summarising academic literature. ChatGPT emerged as the most widely adopted tool, with 86.2% of students who use GenAI selecting

it as their tool of choice. The widespread adoption of GenAI highlights its growing prominence and underscores the importance for educators in higher education to be knowledgeable about the capabilities of tools like ChatGPT (Almassaad et al., 2024). Understanding students' willingness to use—and concerns about—GenAI can help educators better integrate these technologies into the learning process, ensuring they complement and enhance traditional teaching methods.

The current study aims to explore the perspectives and experiences of final-year undergraduate students and MSc conversion students regarding the integration of GenAI. The research questions the current study aims to explore are:

- *How do university students perceive the role of GenAI in their studies, and how do they incorporate it into their learning?*
- *How do students engage with generative AI in their learning, and what risks, needs, and benefits do they identify?*

2 Method

This study used inductive *Reflexive* Thematic Analysis situated within an interpretivist paradigm which recognises researcher and participant subjectivity (Braun & Clarke, 2022). Our research adopted a critical realist ontology (Maxwell, 2012), using student led focus groups to capture the situated, interpreted realities of students' experiences and perceptions of using Gen AI. This approach facilitated sensitivity to students' personal, social, cultural and educational contexts, and how these contexts can shape their approach to and engagement with GenAI tools.

2.1 Participants

Using an opportunity, snowball-sampling strategy, psychology students recruited 29 psychology student participants. The majority of the participants were MSc conversion students (this is a UK-specific one-year MSc programme, which provides students from non-psychology backgrounds with a degree that is accredited by the British Psychological Society). Some of the participants were undergraduate students (see table below for participant backgrounds). Students were recruited from Queen's University Belfast (QUB) and the University of Glasgow (UofG). Six focus groups (FGs) were conducted with a total of 13 participants (12 females and 1 male) from QUB and 16 participants from UofG. The sample was composed of local and international students. Participants were aged 18 or older and were enrolled as students at their respective institutions at the time of the study. Research ethics approval was granted by the Queen's University Belfast Engineering and Physical Sciences Faculty Research Ethics Committee.

2.2 Data collection

A semi-structured interview guide was developed through discussions in the research group. The team at QUB developed the initial interview guide, which was then fur-

ther refined by the team at UofG (e.g., making changes to the way some questions were worded). A final version was agreed upon between the two teams, which was the one used by the students. The interview guide included questions about how students used GenAI for their studies and what their perceptions were about the usefulness and limitations of these tools for their studies. The guide was piloted to test the questions and ensure smooth communication between the interviewees. The questions aimed to capture how students were using these tools for their studies and assignments, their views on GenAI's role in learning and any concerns they had about using them (see interview guide in Appendix B).

All focus groups were conducted online between May and July 2024 by students, with one acting as the moderator and the other as an assistant, who took field notes and asked clarifying questions at the end of the discussions. Each FGD took approximately one hour, with engaging conversations and a friendly atmosphere. These sessions used semi-structured interview questions to maintain consistency while allowing for fluid and natural conversation. In using this qualitative method, we aimed to create a comfortable setting for students to discuss their diverse perspectives candidly (Gill et al., 2008). Participants were given the option to select a pseudonym; those who did are referred to accordingly, while pseudonyms were assigned to those who did not.

The use of GenAI in academic learning and assessment remains a controversial topic. To ensure that participants could openly express their views on the use of such tools, they were recruited and interviewed by two students. Given the nature of the topic, an insider perspective that would allow participants to feel safe and be candid about the use of GenAI tools was essential. Researcher students were trained by staff to facilitate the discussions and ensure that all participants took part in the discussions. Focus group discussions were digitally recorded and transcribed verbatim.

2.3 Analytical approach

Data were analysed inductively by four researchers: one academic staff member (lecturer in HE) and three student researchers. The analysts (Authors 1, X, X and X) first familiarised themselves with the recorded data and conducted multiple readings of the transcripts. Each researcher independently coded the transcripts, followed by group discussions to compare interpretations and deepen the understanding of the data from both student and lecturer perspectives. To explore patterns of meaning across the dataset, we clustered together potentially connected codes until a number of working, provisional themes were developed by individual researchers. In developing themes, we focused on patterns of shared meaning that captured significance in relation to the research questions, prioritizing depth and interpretative insight.

Recognising that lecturers and students may hold different views on the use of GenAI, this collaborative approach allowed for sensitivity to context by exploring alternative interpretations of the data (Yardley, 2008). Discussions between student and staff researchers helped challenge interpretations and surface different understandings, especially regarding power dynamics or insider/outsider perspectives. We focused on identifying contradictions in data and seeking insights into emerging complexities that can result in data that grows both in depth and breadth (Cena et

al., 2024; Yardley, 2008). The process of theme development was documented, and after researchers shared their interpretations of data, initial themes were discussed, and any disagreements were resolved through further discussions to ensure that each theme aligned with the research questions. To ensure quality in research, we followed the criteria for good *reflexive* thematic analysis (Braun & Clarke, 2013). To enhance transparency, an example of staff and student researcher notes and discussions during the analytical process is provided in Appendix B.

2.4 Findings

We identified three themes within the data that capture students' perceptions and experiences of using GenAI at the time the data were collected. These themes highlight the importance students place on using GenAI for their learning, with a focus on emphasising their endorsement of these tools, which are becoming increasingly popular among students, as well as the need to learn how to use them appropriately. The themes discussed below, as shown in Fig. 1, are: (1) *GenAI in Student Life: A Necessary Innovation or a Shortcut?* (2) *GenAI as a 'double-edged sword'*; and (3) *Ethical and Acceptable Use of AI*. The first theme addresses the first research question, whereas the second and third theme addresses the second research question.

RQ1 How do university students perceive the role of GenAI in their studies, and how do they incorporate it into their learning?

1. GenAI in Student Life: A Necessary Innovation or a Shortcut?

This theme explored how students used GenAI as a tool to support their learning at the time the data were collected. The data suggest the growing recognition among all students of the transformative potential of GAI tools in educational environments.

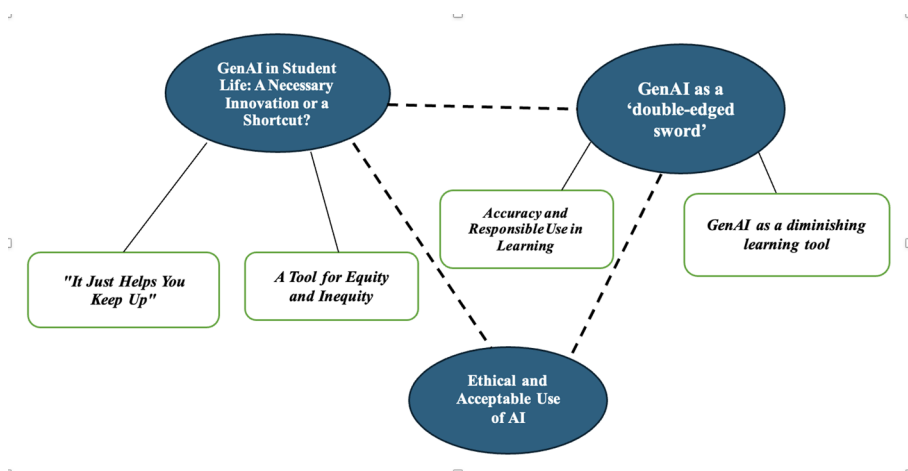


Fig. 1 Thematic Map

It highlights significant advantages of its use, particularly when students feel under pressure, as well as its ability to promote inclusive education.

“It Just Helps You Keep Up” – GenAI as a Study Aid Under Academic Pressure One main reason that students use GenAI is the time pressure associated with their studies. They highlighted that their MSc programs are hectic and require them to perform well and quickly, which drives them towards using AI tools that facilitate fast learning and help them keep up with the reading load of an accelerated program.

I am so stressed that it's really difficult to be thinking critically in some situations and I feel the pressure which I've talked to other students about as well to use tools like this again to outsource some of that just in order to keep up with the quota of work that you have to be producing.—Cory [UoG, FG2].

According to them, GenAI tools ‘save time’ and help explain complex topics in layman’s terms, making them more accessible in a short amount of time. In most cases, these tools are used as study aids to practice learning and provide clarification when students don’t fully understand something from lectures or reading, especially in subjects with constantly evolving information or ‘dense’ content that is difficult to get through. They also appear to be widely used for revision by summarising readings, creating prompts to assess learning, and helping students understand difficult concepts. For students who learn *R* at this level of study, GenAI is used to troubleshoot errors with *R*. In this context, such tools were found helpful for finding potential solutions, especially when students struggled to understand why something didn’t work rather than why it did.

Concerning the use of GenAI for assessments, several students reported finding it helpful to avoid the ‘blank page’ challenge, referring to the overwhelming feeling of starting work from scratch. As two of the students described:

I feel like one of the hardest things about writing frequently is looking at a blank page and being like overwhelmed by the quantity that you need to write. And then, almost like the writing process in a lot of ways is the process of editing and revising. – Aisling [QUB, FG5].

I would sort of input a bunch of information into ChatGPT or whatever, and then get like a really bad first draft, but then be able to work with that as a starting point instead of a blank page and almost use it as an outline. – Clodagh [QUB, FG 6].

As indicated above, students value the draft they get from these tools as a scaffold or outline, a way to get past the initial hurdle and start shaping their ideas. From all the participants, there was no indication of using AI output as final products, but as a product to lower the barrier of writing.

A Tool for Equity and Inequity Findings highlighted the complexities surrounding GenAI as both a tool for equity and a potential driver for inequity in learning.

According to students, GenAI provides a lot of support for neurodivergent and ESL learners. Students with learning difficulties (dyslexia or ADHD) benefit from GenAI in getting explanations in simpler language, making content clearer and more manageable and editing work.

I mean in my own experience and a few other people I've talked to at the uni as well, but with ADHD, it's, it's been very like transformative for a lot of people, particularly in an educational context, but also in like a workplace context or even just their daily lives. - Cory [UoG, FG3].

Whereas, for international students who are non-native English speakers, data suggest that GenAI functioned as a linguistic support tool, helping refine grammar and expression in ways that may reduce language-related anxiety. *For international students, I think one thing that's really important- uh one thing that's really useful um, about recent AI is about the language about language refinement [group nods], grammar and things like that. – Lulu [UoG, FG2].* This is an important aspect in terms of how GenAI is viewed as a tool that reduces structural language disadvantage by helping students express what they know in a more effective way.

It is also perceived as a useful tool for bridging educational gaps so that students from underprivileged backgrounds can use GenAI to supplement their learning. An example mentioned was the advantage that some students have access to resources, such as parents who can assist with assignments, while others lack this crucial support. Therefore, GenAI is perceived to help level the playing field by providing equitable access to resources and information, especially for those without similar help at home. However, some students highlighted that there is a disparity in who can use the advanced tools and who can't, indicating that the cost of advanced AI tools may disadvantage students from lower socioeconomic backgrounds, emphasising the need for universities to ensure fair access to these tools.

You can't have some people having access to ChatGPT, like the paid version, Grammarly, plus this, that the other, and then other people just having access to ChatGPT that only does 2021. Do you know what I mean? Like that's not the same, that's not equal. So, I guess my point would be, if they are going to endorse something, it needs to be equitable — Maisy [UoG, FG1].

Another important point was the growing concern over those who over-rely on using those tools to write their assessments. As some students are reported to be using GenAI more heavily than others, participants believe that this gives certain students an unfair advantage over those who put in the effort themselves. *“Imagine you're trying so hard in class and you're, like, killing yourself for good marks. And then somebody shows up and they're like, this is an essay that a computer wrote.” – Anna [QUB, FG,5].* The frustration expressed by Anna, who feels that students who work hard for their grades are at a disadvantage, underscores the friction between technological advancements and traditional academic expectations.

RQ2 How do students engage with generative AI in their learning, and what risks, needs, and benefits do they identify?

2. GenAI as a ‘double-edged sword’.

The second theme highlights the significant challenges associated with the use of GenAI in education, particularly issues related to the accuracy of information, dependency and critical thinking. This theme is divided into two sub-themes: (1) ‘Accuracy and Responsible Use in Education’ and (2) ‘GenAI as a Diminishing Learning Tool’.

Accuracy and Responsible Use in Learning Questions were raised about the accuracy and reliability of AI-generated content, especially for complex analysis. A common concern was that the tool generates inaccurate information on the topic of inquiry and may produce incorrect content. Specifically, there is a strong concern that AI can generate details that sound convincing but are not factual. In most cases, they emphasised ‘*needing to go back and check requires extra time*’, reflecting that while fast, GenAI is not viewed as reliable, emphasising the need to verify its answers, which can be time-consuming. For this reason, knowing what they were doing in advance was essential for using it effectively, and all students remarked that GenAI output should not be used as a sole source of information. According to them, a ‘*baseline understanding*’ is crucial for getting good results.

Sometimes it gives you things that are not true or it combines two different pieces of information, and you can notice it if you already have that baseline of understanding (...), and if you don't have that kind of understanding and use the information for coursework, that could ruin your entire assignment - Arthur [UoG, FG 1].

If it's helping you, if it's facilitating the learning process and it enriches what they've already learned in class, then it's just like any other learning tool. But if it takes over and does the work for you, then it might become a problem - G [UoG, FG1].

As the above suggests, using these tools to build on prior knowledge is essential, rather than relying on them to complete the work. For them, it is acceptable to use AI as long as they are not relying on it to write the entire assignment or using answers provided by AI without modification.

GenAI as a diminishing learning tool An important point made was the concern that using GenAI consistently can have long-term implications by diminishing learning. They suggested that over-relying on GenAI can lead to ‘cognitive dependency’, where students increasingly outsource problem-solving to AI as a shortcut, ultimately compromising the development of essential long-term learning skills. For example, many students expressed concerns about over-reliance on GAI, which they felt could erode their critical thinking and writing abilities, ultimately leading to a decline in deep learning and intellectual engagement. The ease of accessing it and the ability to

provide quick, ready-made answers made it tempting to bypass the original thought of engaging in the full learning process, ultimately disrupting the creative process. The pressure to complete work quickly and efficiently leaves them little time and tempts them to use these tools, which can lead to relying too heavily on GenAI rather than focusing on developing these learning skills.

There's so much pressure to get things done well and quickly that it sometimes feels like there isn't enough time to develop the important skills we should be learning for our degree. ChatGPT makes it even easier to bypass that learning process, which can be a real concern. – Elaine [UoG, FG 1].

Some argue that using AI for tasks like writing or generating ideas can stifle creativity, as students may not come up with their ideas or learn to think outside the box. For example, one student shared their experience of using AI to generate cover letters for job applications, which led to them not retaining important details or understanding the job requirements fully. They acknowledge that while AI can assist, it doesn't necessarily promote deeper learning or retention, as they would need to rely on AI repeatedly rather than internalising the knowledge themselves.

I sometimes use AI to generate an outline for my assignments by providing bullet points, which I find helpful. However, I don't find ChatGPT's output very useful overall—it's often vague, overly wordy, and lacks specificity. It doesn't replace the critical thinking needed for an assignment. - Iona [UoG, FG2].

Relying on GenAI output to complete assignments, without developing their writing and research skills, is seen as problematic: “If AI is doing the thinking for you, you are not learning”, Sonia (UoG, FG3). Therefore, they indicated that GenAI should be used with a lot of caution for assessments and that it is more helpful for studying than for assessments. Conversely, other students argue that GenAI tools do not negatively impact their critical thinking skills. They believe that due to GenAI's inherent inaccuracies, students still need a fundamental understanding before using these tools effectively and that GenAI should be used as a supplementary tool rather than a replacement for deep learning and critical analysis: “If you don't understand the topic and you're just using that only like it's going to give you faulty information and it's not going to be good, But if you like to blend what you know with it I find that's amazing” Anna – [QUB, FG5].

3. Ethical and Acceptable Use of AI.

Findings highlighted the tension students experience between the need to upgrade their skills and learn to use GenAI and the dilemma of how to use it properly. One of the main concerns remains the lack of clarity around how to use it for assessments, particularly due to fears of plagiarism. They worry that they may unknowingly use GenAI in ways that could be considered cheating or academic dishonesty. The confusion stems from a lack of clarity about how others are using these tools and the

absence of clear boundaries between acceptable and excessive use, leaving students uncertain about what constitutes plagiarism when using such technologies.

You don't know how much other people are using this technology. You don't know what is, like, an appropriate amount. You don't know what plagiarism is, and what is not. That's what like Cory was saying is like, where's the line? It's a personal judgement. - Iona [UoG, FG 2].

Academic integrity issues are especially prominent in group work, where students receive the same mark for group coursework. This stress is amplified when some students submit work believed to be written by GenAI tools. Beyond the perceived unfairness faced by students who put genuine effort into their coursework, the use of AI in group assessments is described as particularly disheartening, as the inclusion of AI-generated work may affect the overall group grade and undermine the contributions of those who completed their work authentically. Concerning assessments, another concern was students' perceptions of lecturers using GenAI for assessment and providing feedback. They express a clear preference for their work to be marked by lecturers and feedback to reflect genuine engagement with their work, not something GenAI generated. They associate human feedback with care, recognition of effort and emotional reciprocity. Statements such as "...I would hope that they're, you know, reading the material and giving you like specific feedback... - Ramina [QUB, FG4] " or "I've done all this work... You should be giving me your personal time now... Aoife [QUB, FG5]" indicates that if staff use GenAI to reduce their own work with marking, they perceive this as an unfair thing to the efforts they put into their coursework.

Students suggested incorporating AI-related skills into the curriculum to prepare them for future job markets. According to them, there is a '*defensive criticism*' of GenAI within academia, reflecting resistance towards technological advancements and concerns about their impact on learning and future employability opportunities.

The resistance of people to learning about AI as they fear that AI will replace us in jobs. However, they think we are very far behind where we are frightened that it is. We shouldn't miss the opportunity of something that will become ubiquitous. - Caoimhe [QUB, FG 4].

It's an amazing part of technology that you know has been taking time to develop, and I think for us to just say, oh, it's all bad, would be naive and it's not for us to say. - Ayne [QUB, FG 5].

GenAI is already prevalent and unavoidable, and thus, they emphasised that it is the responsibility of the HE institutions to teach students how to engage with these '*new sources of learning*'. As a result, the university's stance on GenAI and guidance on its appropriate use are seen as crucial. Students argue that the fear among academic staff should not discourage using those tools, and that academic dishonesty has always existed and that GenAI is simply another potential tool for misuse, rather than the

cause “Dishonest working like this... (generative AI) hasn’t created that. It’s always been there. It’s just maybe a different opportunity to do it.” Anna [QUB, FG6].

Students emphasised that the mindset around GenAI needs to evolve. Since universities are meant to prepare students for the world beyond, and AI is set to play a significant role in the future workforce, they are eager to learn more about it and develop the skills necessary to use it effectively.

It’s a fact that it’s just growing, and it is going to take over the majority of the industries around us. And I think the only right way is to know how to use it and like to maximise the kind of efficiency. So yes, it would be helpful to kind of train people on how they could use it and you know, get the most out of it.

- Farah [QUB, FG6]

Participants highlighted that they didn’t receive any formal training on using AI technology, and the fear of using it inappropriately stems from the fact that different disciplines use AI for different purposes: “I don’t think anyone has had any real training on it, but I think there’s definitely a better way. You sort of work out ways to do things that are more efficient, but not having guidance is a concern.” — Arthur [UoG, FG2].

According to them, universities need to set boundaries on the use of AI and also instruct on how to use it effectively. Lecturers, as educators, are perceived as crucial in the process of setting parameters for acceptable AI use. They asserted that if lecturers are using GenAI, they should discuss how they are using it and be more willing to create clearer guidelines on how students are expected to use it. For example, students believed it was widely understood that GenAI should not be used to complete entire assignments, yet the lack of clarity around the extent to which GenAI is permitted is a source of frustration for them. Given the variety of GenAI tools, students believe universities should explicitly specify which types of GenAI are appropriate for different tasks and adopt a progressive approach to GenAI by developing policies that promote the ethical use of these tools. To support this, students suggested several forms of guidance, including introductory sessions at the start of their studies, releasing videos or short clips, blog posts, training sessions, detailed guides, and establishing an AI support hub with regular updates. They held the belief that such guidelines would benefit not only themselves but also academic staff, who need to assess whether the work is generated by students or GenAI. “If SLd could release even just little clips, you know that you could erm or like a little kind of blog post or something, you know, every so often, that’d be really useful as well as, you know, the more standardised guidance - Heather [UoG, FG2].

3 Discussion and conclusion

The aim of this study was to understand how students use GenAI technologies in higher education and the perceptions of the benefits and challenges of using such tools to enhance their learning. Our study provides important insights into a highly

relevant yet underexplored area at the intersection of higher education and technology. By examining student perceptions of the benefits and challenges of using GenAI in their studies, we gain valuable insights into how these tools are influencing learning and HE. Our findings highlight the opportunities and challenges GenAI presents for students.

Students recognise the growing role of GenAI in education and future careers. They are familiar with and use the tools to aid with their studies and assessments, particularly in demanding programs at the postgraduate level. AI tools provide a convenient solution to this, considering academic pressure and time constraints. Some of the ways in which they use GenAI are to summarise readings, clarify difficult concepts and troubleshooting errors in subjects like R programming. This is in support of previous literature suggesting that familiarity with using GenAI tools and the ease of completing tasks and solving problems make such tools convenient and easy to use (Yang & Zhang, 2024). This can also enhance their academic performance in the short term (Rahman & Watanobe, 2023). In the context of writing assessments, GenAI appears to help students overcome the ‘blank page’ problem by reducing cognitive load through generating an initial draft and lowering anxiety by providing a starting point.

One of the most significant advantages of GenAI is its ability to promote inclusive education by providing substantial support for neurodiverse learners, ESA learners, and non-English speaking students. These findings support previous research suggesting that such innovative systems are capable of tailoring educational content and methods to enable personalised learning and delivery (Chen et al., 2020; Roshanaei et al., 2023). From an equity perspective, our findings suggest that GenAI functioned as tools that helped mitigate structural disadvantages and reduce students’ anxiety. Nevertheless, GenAI may also exacerbate inequities, as the cost of advanced AI tools could disadvantage students from lower socioeconomic backgrounds or cultural backgrounds, highlighting its potential to amplify existing inequalities. This reflects concerns raised by other research suggesting that GenAI’s statistical bias toward dominant knowledge systems and worldviews reinforces existing inequalities (Bender et al., 2021), further marginalising those whose voices are already underrepresented. These findings emphasise the need for universities to ensure fair access to these tools. Injustice may also arise when some students use GenAI more extensively than others, creating an uneven playing field where those who rely heavily on AI can achieve higher marks with less effort. This raises concerns about equity and academic integrity, as students fear that such overreliance could devalue genuine learning and hard work. Our findings resonate with other research suggesting that the use of AI tools can lead to grade inflation for some, while others continue to invest legitimate effort in their studies (Foster, 2024). Findings also indicate that student motivation and learning may be negatively impacted when hardworking students feel disheartened and perceive a sense of injustice, as the distinction between genuine effort and AI-generated work becomes blurred. These findings align with recent work suggesting that such disparities can demotivate students who work hard, and that poses a threat to the overall academic integrity (Davidson, 2023).

Findings revealed several concerns surrounding the use of GenAI in academic settings. A primary issue is its potential to hinder authentic learning, critical thinking, and creativity. Many students reported a lack of clarity about how to appropriately use GenAI, particularly in relation to assignments. While some saw value in using the tool to support their understanding of content, there was widespread caution about over-reliance. This hesitation is grounded in concerns about the lack of accuracy in information, the risk of distorting facts, details that sound convincing but are not factual, and issues that align with research suggesting that GenAI can introduce bias and reduce the accuracy of information (Yang & Zhang, 2024). Findings indicated that without foundational knowledge gained through recommended readings, students may struggle to evaluate AI-generated content effectively, finding it confusing or misleading.

The ease of access to GenAI tools, combined with academic pressure, may tempt students to depend on them excessively. This reliance can disrupt the development of essential academic skills such as critical thinking, problem-solving, and the creative application of knowledge. As recent literature notes, GenAI can provide quick and effortless content that often bypasses meaningful engagement with learning materials (Kostanek & Li, 2025; Thorp, 2023). Consequently, students may develop a false sense of mastery, mistaking tool proficiency for genuine understanding (Brown & Smith, 2024). To address these risks, students must be made aware of the long-term consequences of bypassing traditional learning processes and promoting a balanced approach where GenAI is used as a support tool rather than a substitute can help mitigate these issues. Further, educators can employ familiar assessment pedagogical strategies that develop student evaluative judgment, but adapted to GenAI. According to Bearman et al. (2024), this can be done in three ways: by allowing the learner to assess GenAI outputs; the learner developing their own GenAI processes, by assessing the quality of the prompts used to interrogate ChatGPT; and finally, employing GenAI to assess their own evaluative judgment.

Students are increasingly aware of both the opportunities and risks that GenAI poses to their academic development and future employability. While many recognise its potential as an integral part of modern life, concerns remain about its ethical use, particularly in relation to plagiarism and academic integrity. A strong appetite exists among students to learn how to use GenAI responsibly, yet many expressed uncertainty about what constitutes acceptable versus excessive use. This lack of clarity generates anxiety and confusion, especially in disciplines like psychology, where original thinking and deep theoretical engagement are essential. Findings challenge previous research suggesting students do not view GenAI use as unethical (Consuegra-Fernández et al., 2024). In contrast, students in the current study expressed ethical concerns but struggled to determine the boundaries of appropriate use. These challenges are amplified in collaborative settings, such as group work, where reliance on GenAI by one member may affect the fairness of shared assessments. The fear of plagiarism and the erosion of originality in student work were consistent themes across the dataset. These concerns align with broader literature highlighting the risks of over-reliance on language models, which lack criti-

cal thinking, subject-specific understanding, and the capacity to generate original academic insights (Liu et al., 2023; Weidmann, 2024). Without structured policies and targeted training, students risk becoming overly dependent on GenAI tools, potentially compromising their intellectual development. To mitigate these risks, students clearly advocated for the development of standardised university guidelines, including discipline-specific examples and accessible training on ethical and effective use.

3.1 Limitations

This study has some potential limitations. First, the data were collected in mid-2024, and given that the GenAI landscape continues to evolve rapidly, student perceptions of these tools may change over time. Second, most of the data were collected from postgraduate students and only a few undergraduate students, which limits the ability to understand the perceptions of undergraduate students in the use of AI tools. Third, the sample in this study is gender imbalanced, with more females than males. However, this reflects the wider demographics of psychology students, with women comprising approximately three-quarters of undergraduate and postgraduate cohorts (Odic & Wojsik, 2020). Fourth, universities have adopted GenAI policies to varying degrees, providing different levels of guidance - for example, some prohibit its use in all assessments, while others allow it if acknowledged (for a non-exhaustive list of UK policies, see Barker, 2025). Based on the findings, this variation can contribute directly to uncertainty and anxiety about what constitutes acceptable use, particularly in assessment and group work contexts. Rather than reflecting individual misunderstanding, these experiences were shaped by navigating inconsistent or ambiguous institutional expectations. As a result, students were left to rely on personal judgment, which heightened fears of inadvertent plagiarism and ethical misconduct. Therefore, the conclusions of this study should be interpreted with caution. While insight was provided into the uses and challenges of what students perceive regarding AI technology, the impact of whether a cause-effect relationship exists between student usage of AI and academic performance can only be drawn out through experimental designs.

3.2 Implications

These findings have significant implications for academic institutions. As AI technology advances, educational institutions must also adapt their strategies to protect the integrity of academic achievements and uphold the high standards that form the foundation of higher education. The use of GenAI presents challenges for students and educators, particularly in terms of its potential impact on student learning skills. Students may not always be aware of how their reliance on such tools can affect their development of these skills, which could have consequences for their future. Additionally, the use of technology to write assignments creates challenges in identifying how and where these tools are used, especially when students do not explicitly or honestly disclose their use, and when no detection tools are available to identify such

instances (Kostanek & Li, 2025). This uncertainty makes it difficult to evaluate performance, track progress, and provide meaningful feedback, ultimately complicating the assessment process and the support of student learning.

HE institutions should consider adopting the use of GenAI in a more systematic way that is useful for students for their studies, as well as preparing them for future employment. This involves educating students on how to effectively use GenAI in their academic journey, including the benefits and limitations of these tools, as well as what constitutes acceptable and unacceptable use. Since the time data were collected, various HE institutions have worked hard to develop student guides on how to use GenAI and understand its potential in an ethical way. There should be information on the importance of lifelong learning skills, which are important for future employability, and cannot be replaced or learned otherwise. Such information should be formalised with clear guidelines to help students understand the boundaries of such tools and use them responsibly. Emphasis should be given to the long-term effects that engaging in the misuse of AI tools can have on essential learning skills gained at the Higher education level.

Enhancing early AI literacy across all programmes allows students to learn about the usefulness and limitations of tools as well as responsible use that does not compromise student learning and academic integrity (Kostanek & Li, 2025). One way to achieve this would be designing assessments that preserve the development of essential cognitive skills and reflect authentic learning rather than work generated from AI tools. An important consideration is to design assessments that preserve essential traditional learning skills, that focus on higher-order thinking skills. This means developing assessments that require students to brainstorm ideas using GenAI but do not write the assignment for them (e.g., assessments that require students to critically evaluate GenAI output (Bearman et al., 2024). Development of assessments and best practices collaboratively with students can be useful in understanding the types of assignments and pedagogical approaches that ensure the ethical use of AI.

Finally, the study reveals a complex landscape where the potential benefits of GenAI in education are counterbalanced by significant concerns from students. It is clear that GenAI tools could enhance learning, promote inclusivity, and reduce student workload. However, these advantages must be weighed against the risks of dependency, academic dishonesty, and the decline of learning skills. These findings highlight that the swift evolution of AI technologies has found universities unprepared and uncertain about their acceptability and the potential AI has in supporting student learning and adapting to a changing employment landscape because of such technologies. High exposure to AI integration in universities is unavoidable. AI is already widely used by students. Therefore, HE institutions need to adopt a balanced approach towards integrating AI technologies into higher education. Our evidence underscores the ongoing and future transformation of academia in response to these technological changes.

Appendix A

Table 1 Participant Information

UoG Participants	Age	Gender	Student Type		
Megan	33	female	PGT - MSc Conversion	International	Mature student
Heather*	25	Female	PGT - MSc Conversion	Home (UK)	Non-mature student
Arthur Ignacius	29	Male		Home (UK)	Mature student
Sonia	26	Female	PGT - MSc Conversion	Home (UK)	Mature student
Lauren*	21	Female	UG - 3rd year	Home (UK)	Non-mature student
Elaine*	26	Female	UG - 3rd year	Home (UK)	Mature student
Maisy*	20	Female/woman	UG - 3rd year undergraduate	Home (UK)	Non-mature student
Cory*	32	Nonbinary	PGT - MSc Conversion	International	Mature student
G	22	female	UG - 4th year	International	Non-mature student
Dana*	21	female	UG - 3rd year	Home (UK)	Non-mature student
Rachel*	24	Female	PGT - MSc Conversion	International	Non-mature student
Hannah*	22	Female	PGT - MSc Conversion	International	Mature student
Helena*	21	Female	UG - 3rd year	Home (UK)	Non-mature student
Lulu	27	Female	PGT - MSc Conversion	International	Mature student
Iona	22	Female	UG - 3rd year	Home (UK)	Non-mature student
QUB Participants	Age	Gender	Student Type		
Aisling	25	Female	PGT - MSc Conversion	Home (UK)	Non-mature student
Ramina	23	Female	PGT - MSc Conversion	International	Non-mature student
Clodagh	32	Female	PGT - MSc Conversion	Home (UK)	Mature student
Ayne	28	Female	PGT - MSc Conversion	Home (UK)	Mature student
Nora	26	Female	PGT - MSc Conversion	Home (UK)	Mature student
Riley	30	Female	PGT - MSc Conversion	Home (UK)	Mature student
Cara	26	Female	PGT - MSc Conversion	Home (UK)	Mature student
Anna	27	Female	PGT - MSc Conversion	International	Mature student
Caoimhe	27	Female	PGT - MSc Conversion	Home (UK)	Mature student
Farah	25	Female	PGT - MSc Conversion	International	Non-mature student
Aoife	28	Female	PGT - MSc Conversion	Home (UK)	Mature student
Saoirse	26	Female	PGT - MSc Conversion	Home (UK)	Mature Student
Kabir	24	Male	PGT - MSc Conversion	International	Non-mature student

Appendix B

Interview Guide for Focus Groups with Students

Hello and thank you for agreeing to participate in this interview. The purpose of this study is to understand your views in the use of Generative AI in higher education in Psychology. Your insights will provide valuable information about the impact of AI on your academic journey. Please note that your responses will remain confidential and will only be used for research purposes. Please avoid disclosing identifiable information about students/staff members. The researchers would like to remind all participants to respect the privacy of fellow participants and not repeat what is said in the focus group to others. Equally, it is important that the information discussed during this focus group is not discussed outside of the room. We also request that during the interview participants respect each other's opinions and to have care in the use of language. Using discriminatory or derogatory language will result in exclusion.

- Introductions (participants – their names).
- Check that it is okay to record the session.
- Ask if they have any questions.

As stated previously, this project aims to provide a comprehensive understanding of your views in the use of Generative AI or ChatGPT in higher education in Psychology and how it can be used. Do you think students should use AI for their studying or assignments?

1. Have you ever used AI tools or applications to support your learning in psychology courses during your time at university?
 - a If yes, could you tell us why you have used them?
 - b If not, why have you not used them?
2. What have you used it for (e.g. doing assignments, making summaries of the lecture?) and did you find it helpful for your learning for these things?
 - a If you have found it helpful, what things has it been more helpful for and why?
 - b If you haven't found that it's helpful, why do you think not?
3. What are the main concerns you have about using AI in your studies or assignments?
4. From your perspective or experience, how can you use AI in a safe way to support your learning?
5. Do you think lecturers should integrate AI in their teaching and assessment? If yes, in what way? If not, why do you think it should not be integrated?
6. How do you think university teaching staff could help students understand the appropriate and safe ways of using AI?
 - a For example, would you find guides helpful or should there be training sessions to guide students through the steps of does and don'ts in using AI?
 - b Here, you could also discuss any examples of guidance to AI that you've seen or been given and how/why you found them helpful?
7. Ask if they would like to add anything else to the discussion.

Appendix C

Additional information about the analysis process

Student and staff researchers made notes on the transcripts independently, then discussed their analytic observations and insights, noting any contradictions. This informed development of themes.

Example of how staff and students recorded and discussed their interpretations

Notes from student researcher on transcript 3

- AI can be used for summarizing, revision, coding, grammar and structure, specific googling and breaking down complex information but not complete generation.
- Emphasizing again the importance of integrating AI into learning as a new modern technology.
- Evidence in this one again of AI being used in research techniques, in this case literature reviews and what statistical test to use.
- Again making the point that ChatGPT is not the only AI based software e.g. Grammarly.
- Another point for accuracy of information being a major concern using AI and not having the knowledge to reliably know if its wrong.
- Additional point here that if you did take information from it, what source has that come from and how on earth do you correctly reference it.
- There is the repeated concern here as well that certain words and structures make AI very identifiable to checking programs like Turnitin.
- Repeated concern that you're not learning as much, with an extra extension that it decreases your creativity and you become over reliant on it. Linked to the T2 mention that the point of your degree is to learn not use AI and therefore how on earth are you going to get a job.
- Repeated belief the smaller efficiency tasks are much safer and reliable than finding new information or generation.
- Repeated point that different universities have extremely varying AI policies which can make students wary to use it with a repeated request for guidance. Any guidance that is given is usually a general university policy that is too vague to the specific student needs to be of any use.
- Some staff have given examples of how and when to use AI and this is very appreciated for setting guidelines as well as raising awareness for the different ways it can be used. Alongside guidance they would enjoy a learning course on how to best use it.
- Point here that clearly integrating it may remove negative stereotypes.

Notes from staff researcher- transcript 3

- Why is it being used
- Summarizing a piece of information
- Generating practice questions
- Using it to write answers
- Get a piece of information you aren't able to get it yourself
- Important that we integrate AI as we have to keep up with the modern technology
- Important for international students, language refinement, grammar
- AI provides comprehensive answers to questions, compared to just searching on Google

- Time efficient (compared to google)
- Takes too long to do literature reviews, now AI programmes can help you do that
- Convenient and saves a lot of time
- Used it in helping with coding, as that was new to students
- AI can guide you in what coding is trying to do
- Too many answers in google, hard to know which one is the right answer
- Used it for word count for dissertation, to help create a template of how many words each section should be
- To summarise lecture slides – get bullet points out of it
- Generating questions related to the teaching content, to help revise
- Grammar, paraphrasing
- Scite for research – helps refine research
- Grammarly for language
- Helped to choose which statistical model to use and to help debug codes.

Fears

- Accuracy of information (a definition, references)
- Not confident enough that the output is correct
- Need to check it More time consuming
- Doesn't give the right credit to people's work
- Not giving the right credit to the ones who have produced the work
- Ai pulls from it and spits it out
- It doesn't have the morals
- When needing to go back and check, requires extra time
- When editing, it uses certain English words quite a lot
- Makes work easily detectable even own students work
- You don't get to use the brain with it
- Hampers creativity
- Realising that AI can impact capacity to get the degree
- Accuracy the main concern

Using it safely/not safely

- Double checking R codes
- Good for revision and practice
- Unsafe when asking it to generate big chunks of information
- Using it blindly not right
- Unsafe to rely on whatever it generates
- Can give you great ideas and push you in the right direction of what form of research is best for you
- Should be used as a frame of reference not to copy paste information from it 'It's not why we are here'

Lecturers are important in this process

- Expectations for lecturers to use it
- Students trust lecturers in how to use AI and what is safe and what is not about it
- What can you do what can you not do – should come from uni
- Clarity is needed about what is acceptable and safe
- If integrated and seen by lecturers use it, it gives confidence to students to know where and how to use it
- Don't know what's the best way to use it
- Students want the information so they can build off that themselves

Example 2: Discussion between the student and staff researcher about their interpretations of the transcript

I have read through the findings and discussion section and your interpretation of the transcripts aligns very highly with the notes I had made when viewing the transcripts. I particularly appreciated the emphasis that many of GAI's problems or fears come from overuse of the system or lack of acceptable use guidelines where more general use can be very beneficial to students. AI accuracy issues, use for neurodivergent students, importance of AI in future workplaces and ethical/fairness concerns were all notes I had also made when viewing the transcripts too.

The only additional point I had that is not overly discussed (although I believe still mentioned) was student opinion on staff using AI. I had noted they were generally supportive of this, however were adamantly against staff using AI for marking, believing face to face feedback was still better. Another student had a more vindictive opinion on this, that staff should still take the time to mark in respect to the time it took students to complete assignments. However, this came up in maybe only 1 or 2 transcripts and was more an interesting point that stood out to me rather than a major theme.

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Data availability The data generated and/or analysed during the current study are not publicly available as participants were not informed that their data would be shared, and the ethics documentation was not designed to support data sharing. Therefore, sharing would not comply with the conditions of informed consent or ethical approval.

Code Availability Not applicable

Declarations

Ethical approval Research ethics approval was granted by the Queen's University Belfast Engineering and Physical Sciences Faculty Research Ethics Committee.

Consent for human participants All participants provided informed consent prior to taking part in the study. They were given information about the purpose of the research, what participation would involve, and their right to withdraw at any time without consequence. Consent was obtained in written and verbal form in accordance with ethical guidelines approved by Queen's University Belfast Research Ethics Committee.

Conflict of interest None.

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